

The Ephemeris

June 2018

Volume 29 Number 02 - The Official Publication of the San Jose Astronomical Association



June 2018 - September 2018 Events

Board & General Meetings

Saturday 6/2, 6/30, 7/28, 8/25, 9/22

Board Meetings: 6 -7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 6/3, 7/1, 8/5, 9/2

Solar Observing (locations differ)

Sunday 6/3, 7/1, 8/5

Intro to the Night Sky Class

Houge Park 1Q In-Town Star Party

Friday 6/22, 7/6, 8/17, 9/14

Houge Park 3Q In-Town Star Party

Friday 6/8, 7/20, 8/31,

RCDO Starry Nights Star Party

Saturday 6/9, 7/7, 8/4, 9/1

Imaging SIG Mtg

Tuesday 6/19, 7/17, 8/21, 9/18

Astro Imaging Clinics

(at Coyote Valley OSP)

Saturday

Quick STARt (by appointment)

Friday

Binocular Star Gazing

6/9

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Photo by SJAA Member: **Paul J Mahany**

May 11, 2018 M 88

Location: Sierra Nevada, California

SJAA 's School Star Party Program Needs Your Help!

Article by Rob Jaworski

The San Jose Astronomical Association is an educational, science based non profit organization that has been operating in the south bay for over sixty years. Its mission is to bring the science of astronomy to the public by providing many different events and resources, most of which are free and open for anyone to experience. SJAA members set up telescopes at public parks for anyone to step up and get a close-up view the Moon or Jupiter. Astronomy talks, "Getting Started" classes and even telescope loans are examples of what the SJAA offers.

It's been said that astronomy is both a humbling and character building experience. In this age of hubris and arrogance, in today's world of screens both small and large, people, especially young people, may find it difficult to find examples of "humble". They may not know what it means to build character. But there is hope! The SJAA's School Star Party program partners with dozens of schools each year and touches hundreds of young people, students and their families and introduces them to the glorious sights of the night sky, its stars, constellations, planets and more. It's where people regularly see Saturn's rings or Earth's crusty, cratered Moon for the first time. They don't believe they are seeing it live, but when told it's the real thing, they shout with exclamations of joy. Every. Single. Time.

After 30 years of organizing and leading the SJAA School Star Party Program, Jim Van Nuland has retired. We now realize it will take a New Team to take over all the program aspects that Jim managed by himself.

SJAA is looking for a number of people who can help with this program. If you have a telescope or binoculars, or can explain very basic information about the night sky, or if you are good with organizing, we would love to hear from you. The School Star Party (SSP) program has several distinct roles that need to be filled, from fielding incoming requests for school star parties, to planning the event, or simply showing up and showing off the night sky. Read about them below.

If you are interested in learning more, please send an email to AskSJAA@sjaa.net.

There are four distinct roles, or areas of responsibility, for the SSP program. They can each be handled individually or two or more can be held by the same person. Here are their descriptions.

SSP Program Leader

After many, many years of leading the School Star Party program, Jim Van Nuland has decided to retire from it. This opens up the possibility for someone to pick up the reigns, modernize the program, and build it to suit today's needs. If you have, or are looking to acquire, leadership skills, please consider stepping into this important role. Responsibilities will be greater in the beginning as you learn about how the program has been operating, the tools and techniques used, and as you work to revitalize the systems. Afterward, the Program Leader role will mainly consist of reporting to the board of directors status of program, number of schools served, contacts made and the like. As Program Leader, you can decide if you'd like to also be the SSP Liaison, SSP Scout, and/or Astronomer. You will have the flexibility to design and run it as you see fit (within the constraints of a reasonable school star party program, of course). Keep in mind that although you will report to the Board on a monthly basis, your attendance at monthly board meetings is not required.

SSP Liaison

This role handles incoming requests for SSPs. When a teacher, parent or other member of a school community contacts the SSP program, the Liaison does the initial reply. In this reply, all the info that the member needs to know is provided. This includes things like how to determine which dates are good to have a star party, which upcoming dates are available and guidelines for having a successful school star party. Then internally, the Liaison starts working to identify a Scout who will 'own' this engagement with this school. Once the community member selects a few possible dates, the Liaison hands off the communication to the Scout with a message introducing her or him.

SSP Scout

The next role is that of the School Star Party (SSP) Scout. The SSP Scout owns the relationship with this community member and this school for the duration. The SSP Scout's duties include finalizing the SSP date out of the possible dates identified by the Liaison, then going to the school on some day before the SSP to survey the site, meeting the member and stakeholders in person, finding out if there are any special needs like insurance, and providing resources to help promote the event. At the same time, the SSP Scout identifies SSP Astronomers who can commit to the event. The Scout leads the event all the way through to the end of the event at that school.

SSP Astronomers

SSP Astronomers are the boots on the ground. They are the ones who get to hear the Oohs and the Ahhs from members of the school community. They are the on the ground liaisons between the SJAA, the community, and the wonders of astronomy. They provide their own scopes and related gear, share their knowledge, and they exercise their skills in clearly communicating to the general public. SSP Astronomers have to be able to commit to the date, communicate with the Scout, and work out their own transportation to and from the SSP, for themselves and their gear. The Scout does not necessarily have to be an SSP, but they do need to be at the event from start to finish, so it makes sense for her or him to bring a scope.

Cupertino Earth Day April 21, 2018 by Bill O'Neil

This year was the 10th annual Cupertino Earth Day Festival. This is the 6th year in a row that SJAA has attended. Thank you to SJAA Members Marion Barker, Paul Colby, Terry Kahl, Kevin Lahey, Paul Nowicki, Wolf Witt and Bill O'Neil, for coming and bringing their solar telescopes.

Wolf set up the new (donated) 80mm Lunt H-a scope, with our Mallincam video camera and bright 19" Outdoor monitor, so that many visitors could see the TV image without queuing up to look through the eyepiece. See Photos attached from Wolf, Marion & Bill.

Marion Barker manned the SJAA Table with Photo Books of pictures from the Imaging SIG Group, plus O'Neil photos from the 2017 Total Solar Eclipse. Swami also stopped by to talk to the public about SJAA, and give out printed Handouts about SJAA Membership and Meetup events.



Bill O'Neil with the NASA Solar Sign



Swami Nigam and Kevin Lahey



Table 81 manned by Marion Barker



Wolf Witt with TV

SJAA Logo-Wear!

Report and photos by Marianne Damon

Show our pride and let the public know who we are! Making your own custom logo-wear is easy and quick. We now have a webpage on our SJAA website where you can:

- Download the SJAA logo file to your computer.
- Select a logo-wear company. SJAA does not endorse any one company. You can use a company you have ordered from before, or find one that has the items you want.
- Select the “Design/Create Your Own” option
- Select the product available for that option
- Follow the instructions for importing the SJAA logo file onto the site
- Follow the instructions for placing the logo on the front & back, sizing the logo, and inserting a text box if you wish to add your name. You can customize the font type and color of text within the text box.
- Save the design, or finish and place in your cart. When finished with all items, go to cart and check out/order.

I’ve road-tested this method and within one week I received my long-sleeved shirt with a pocket logo on the front, where I inserted my first name, and a large logo on the backside. I also made a carry-bag, with the logo and my full name on one side, and just the logo on the other side. Some companies have basic, but adequate, selections of colors and styles, and others have more extensive options. For example, a company might have basic shirts, sweatshirts and baseball caps, and another company might have shirts of different brands, more colors, a variety of hats, and a selection of jackets. You can select the type, color, quality and pricing of your choosing.

We are looking forward to seeing more of our logo at club events! I remember a young boy and his mom I walked over to in the parking lot. They were wondering if they were at the right place for one of our meetings: “Mom, she’s wearing an Orion Nebula t-shirt, I think we’re in the right place”...had I been wearing my SJAA shirt, there would have been no doubt at all!



*SJAA Logo Book Bag
Photo courtesy of Marianne Damon*



*SJAA Logo Book Bag with Name
Photo Courtesy of Marianne Damon*



*SJAA Logo Shirt with Name
Photo Courtesy by Marianne Damon*



*SJAA Logo Shirt Back
Photo Courtesy by Marianne Damon*

What in the World is an ‘Exoplanet?’



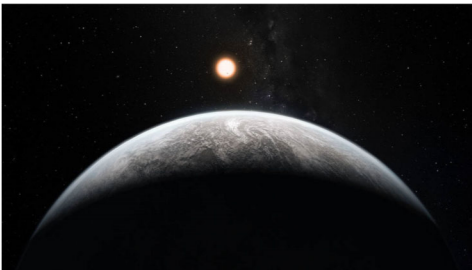
The Milky Way, our own galaxy, stretches across the sky above the La Silla telescope in Chile. Hidden inside our own galaxy are trillions of planets, most waiting to be found. **Credits: ESO/S. Brunier**

Step outside on a clear night, and you can be sure of something our ancestors could only imagine: Every star you see likely plays host to at least one planet.

The worlds orbiting other stars are called “exoplanets,” and they come in a wide variety of sizes, from gas giants larger than Jupiter to small, rocky planets about as big around as Earth or Mars. They can be hot enough to boil metal or locked in deep freeze. They can orbit their stars so tightly that a “year” lasts only a few days; they can orbit two suns at once. Some exoplanets are sunless rogues, wandering through the galaxy in permanent darkness.

That galaxy, the Milky Way, is the thick stream of stars that cuts across the sky on the darkest, clearest nights. Its spiraling expanse probably contains about 400 billion stars, our Sun among them. And if each of those stars has not just one planet, but, like ours, a whole system of them, then the number of planets in the galaxy is truly astronomical: We’re already heading into the trillions.

We humans have been speculating about such possibilities for thousands of years, but ours is the first generation to know, with certainty, that exoplanets are really out there. In fact, way out there. Our nearest neighboring star, Proxima Centauri, was recently found to possess at least one planet – probably a rocky one. It’s 4.5 light-years away – more than 25 trillion miles (40 trillion kilometers). The bulk of exoplanets found so far are hundreds or thousands of light-years away.



The bad news: As yet we have no way to reach them, and won’t be leaving footprints on them anytime soon. The good news: We can look in on them, take their temperatures, taste their atmospheres and, perhaps one day soon, detect signs of life that might be hidden in pixels of light captured from these dim, distant worlds.

The first exoplanet to burst upon the world stage was 51 Pegasi b, a “hot Jupiter” 50 light-years away that is locked in a four-day orbit around its star. The watershed year was 1995. All

This rocky super-Earth is an illustration of the type of planets future telescopes, like TESS and James Webb, hope to find outside our solar system.

Credits: ESO/M. Kornmesser

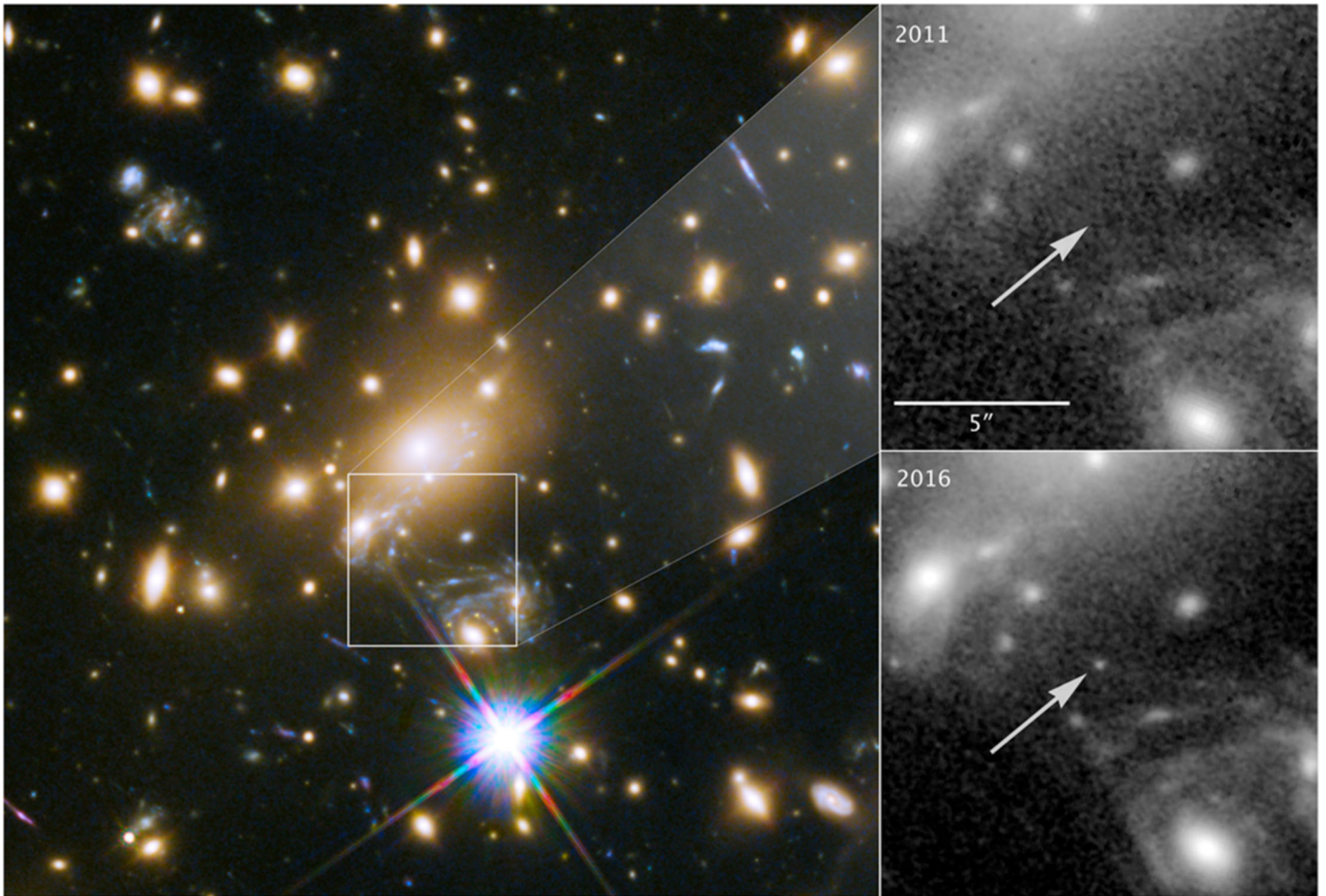
But a few hints had already emerged. A planet now known as Tadmor was detected in 1988, though the discovery was withdrawn in 1992. Ten years later, more and better data showed definitively that it was really there after all.

And a system of three “pulsar planets” also had been detected, beginning in 1992. These planets orbit a pulsar some 2,300 light-years away. Pulsars are the high-density, rapidly spinning corpses of dead stars, raking any planets in orbit around them with searing lances of radiation.

Now we live in a universe of exoplanets. The count of confirmed planets is 3,700, and rising. That’s from only a small sampling of the galaxy as a whole. The count could rise to the tens of thousands within a decade, as we increase the number, and observing power, of robotic telescopes lofted into space.

Credit: nasa.gov

Hubble Uncovers the Farthest Star Ever Seen



Icarus, whose official name is MACS J1149+2223 Lensed Star 1, is the farthest individual star ever seen. It is only visible because it is being magnified by the gravity of a massive galaxy cluster, located about 5 billion light-years from Earth. Called MACS J1149+2223, this cluster, shown at left, sits between Earth and the galaxy that contains the distant star. The panels at the right show the view in 2011, without Icarus visible, compared with the star's brightening in 2016.

Credits: NASA, ESA, and P. Kelly (University of Minnesota)

More than halfway across the universe, an enormous blue star nicknamed Icarus is the farthest individual star ever seen. Normally, it would be much too faint to view, even with the world's largest telescopes. But through a quirk of nature that tremendously amplifies the star's feeble glow, astronomers using NASA's Hubble Space Telescope were able to pinpoint this faraway star and set a new distance record. They also used Icarus to test one theory of dark matter, and to probe the make-up of a foreground galaxy cluster.

The star, harbored in a very distant spiral galaxy, is so far away that its light has taken 9 billion years to reach Earth. It appears to us as it did when the universe was about 30 percent of its current age.

The discovery of Icarus through gravitational lensing has initiated a new way for astronomers to study individual stars in distant galaxies. These observations provide a rare, detailed look at how stars evolve, especially the most luminous stars.

"This is the first time we're seeing a magnified, individual star," explained former University of California at Berkeley postdoc and study leader Patrick Kelly now of the University of Minnesota, Twin Cities. "You can see individual galaxies out there, but this star is at least 100 times farther away than the next individual star we can study, except for supernova explosions."

Credit: nasa.gov



Imaging telescope or lens: Stellarvue SV4

Imaging camera: ZWO ASI1600MM-Cool

Mount: Astro-Physics Mach1 GTO

Guiding telescope or lens: Stellarvue SV4

Guiding camera: QHY5L-II-M

Focal reducer: Astro-Tech AT2FF Field Flatteners

Software: ProDigital Software StarSpikes Pro 4, PS, PixInsight, Sequence Generator Pro, PHD2



Imaging telescope or lens: GSO RC 6" RC6

Imaging camera: Canon 500D (Mod)

Mount: Orion Sirius EQ-G (HEQ5)

Guiding telescope or lens: Guide Scope William Optics 50x200 F/d:4

Guiding camera: ZWO ASI 120 MC

Software: Stark-Labs PHD-Guiding, PixInsight 1.8 Ripley PixInsight Core 01.08.04.1195 Ripley (x64),



Imaging telescope or lens: TPO 12" Truss RC

Imaging camera: ZWO ASI 1600CM Cooled

Mount: Orion HDX110 EQ-G

Guiding telescope or lens: Orion ST-80

Guiding camera: QHYCCD QHY5L-II mono

Focal reducer: Astro-Physics CCDT67

Software: Topaz Labs DeNoise, Adobe Photoshop CS6, Picasa, CCDStack2

Filter: Astronomik CLS CCD 2"



Photo by Francesco Meschia

May 13, 2018

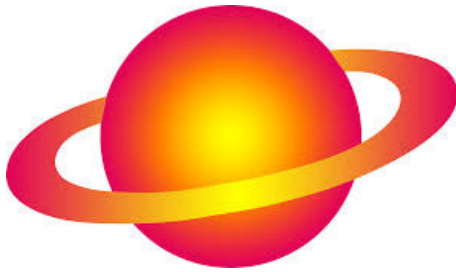
Leo Triplet (M65, M66, NGC3628): 1h54m total integration time (19x360s) at ISO 1600, same camera and telescope.

Kid Spot



Kid Spot Jokes:

- ♦ **How did the astronaut serve drinks?**
In sun glasses!
- ♦ **How many astronomers does it take to change a light bulb?**
Zero. Astronomers aren't afraid of the dark.
- ♦ **What is more useful, moon or the sun?**
Moon because it shines during the night when you need the light; whereas the sun shines during the day when you don't need it.



Kid Spot Quiz:

- 1. The day on which the sun's rays cross the celestial equator is called?**
a) equinox b) solstice c) ecliptic
- 2. Which of these objects is further from the sun?**
a) Neptune b) Kuiper belt c) 90377 Sedna
- 3. When did Space Age begin?**
a) 1942 b) 1989 c) 1957

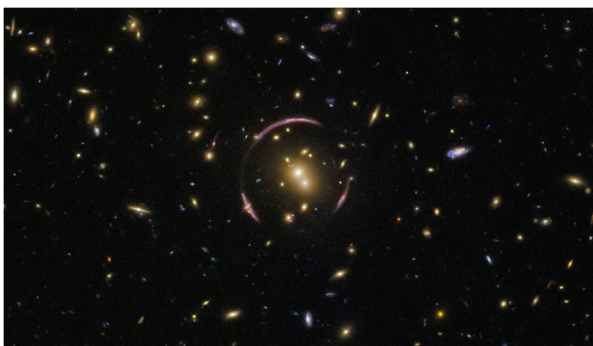
Hubble Finds an Einstein Ring

This image is packed full of galaxies! A keen eye can spot exquisite elliptical galaxies and spectacular spirals, seen at various orientations: edge-on with the plane of the galaxy visible, face-on to show off magnificent spiral arms, and everything in between.

With the charming name of SDSS J0146-0929, this is a galaxy cluster — a monstrous collection of hundreds of galaxies all shackled together in the unyielding grip of gravity. The mass of this galaxy cluster is large enough to severely distort the space-time around it, creating the odd, looping curves that almost encircle the center of the cluster.

These graceful arcs are examples of a cosmic phenomenon known as an Einstein ring. The ring is created as the light from a distant objects, like galaxies, pass by an extremely large mass, like this galaxy cluster. In this image, the light from a background galaxy is diverted and distorted around the massive intervening cluster and forced to travel along many different light paths toward Earth, making it seem as though the galaxy is in several places at once.

Credit: ESA/Hubble & NASA; Acknowledgment: Judy Schmidt
Text: European Space Agency



Kid Spot Quiz Answers:

1. A
2. C
3. C

From the Board of Directors

Announcements

The SJAA is currently looking for a new Ephemeris Content Editor

Board Meeting Excerpts

March 3, 2018

In attendance

Rob Jaworski, Rob Chapman, Wolf Witt, Bill O'Neil, Vini Carter, Swami Nigam, Glenn Newell, Sukhada Palav
Absent Excused: , None
Guest: Jim van Nuland, Emi Nikolov, Nikola Nikolov, Victor Nikolov, Gary, Werner

Officer Elections

Swami Nigam was elected as new President of the SJAA. He is still a board member as well. All other officers will continue in their current roles for another term.

New Ephemeris Production editor elected

Emi Nikolov was voted in as the new Ephemeris Production editor.

Librarian moving away

The current librarian is moving away. Currently looking for someone to fill the position. In the meantime until the position is filled Rob Chapman will be acting as librarian.

March 31, 2018

In attendance

Swami Nigam, Rob Chapman, Rob Jaworski, Vini Carter, Glenn Newell, Sukhada Palav, Wolf Witt
Absent Excused: Bill O'Neil
Guest: Emi Nikolov, Rob T, Jann F

Empty Board Seat

Rob Chapman will send out an announcement by next meeting that we are looking for a replacement for Teruo's empty seat.

SJAA School Program Lead

Jim Van Nuland has announced that he is looking for a replacement leader

for the program. For additional info about this please see Page 2.

Loaner Solar Scope

Wolf Witt asked the board for a proposal for the solar gear for about \$2000. The board approved this.

New welcome email went live

New welcome email went live to new and renewing members, which includes relevant links to club activities

April 28, 2018

In attendance

Sukhada Palav, Bill O'Neil, Glenn Newell, Vini Carter, Rob Jaworski, Rob Chapman, Swami Nigam, Wolf Witt
Absent None
Guest: Emi Nikolov, David Grover, Sarah, Kanch, Jim Van Nuland, Adrianus Zwiitjzen, Marianne Damon

Empty Board Seat

Five people have expressed an interest in serving on the board. Rob Chapman is following up with them

City of San Jose Contract

The City has accepted our RFQ. There are some open follow up items with the city. San Jose wants a signed contract returned in person.

Kelley Park star party

SJAA has been contacted by Keep Coyote Creek Beautiful to conduct a star party following KCCB's Kelley Park Sunset Walk events which run from 7-9pm. The star party will begin at the conclusion of the Sunset Walk about 9pm. KCCB's Sunset Walks .series was very successful. We may move an ITSP from Houge to Kelley Park for the Aug or Sep events.

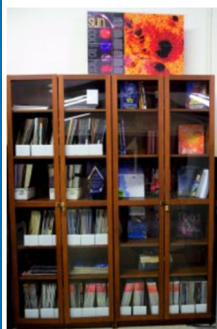
SJAA Logo Wear

Marianne has done some research into making SJAA logo wear available to members. She presented her research and proposal.

Marianne has written up several possible options. The result of her research was that the easiest option for the club would be to provide the logo file on our website, with directions on how to or-

der custom gear with the logo. For additional info on this please see Page 4

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The next Quick Start session during this time period is July 13th.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly

session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association

conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Swami Nigam
Vice Pres/Dir:	Bill O'Neil
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Vini Carter
Director:	Open
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Emi Nikolov
Fix-it Program:	Vini Carter
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Vacant
Loaner Program:	Muditha Kanchana
Memberships:	Dave Ittner
Publicity:	Rob Jaworski
Quick STARt	Dave Ittner
Donations:	Dave Ittner
Solar:	Bill O'Neil
Starry Nights:	Carl Wong
School Events:	Open
Speakers:	Sukhada Palav
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Place
postage
here

San Jose Astronomical Association
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San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with **online** Ephemeris
- ☐ **\$30** Regular Membership with **hardcopy** Ephemeris mailed to below address
- ☐

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____